MQL Calculated Time to Contact (TTC) - Technical Documentation Overview and Implementation

Overview

The MQL Calculated TTC system measures how long it takes sales reps to respond to Marketing Qualified Leads. It leverages LeanData's Time to Action Tracker functionality combined with custom Apex logic to calculate and store response time metrics on Lead and Contact records.

Primary Field: `MQL_Calculated_TTC__c` (Number field on Lead and Contact)

What it measures: Total elapsed seconds from MQL Open until rep takes action (or SLAs fully expire)

Business Logic

The MQL Lifecycle

```
┌──────────────────────────────────────────────────────────────────
────┐
│  MQL Opens          SLA #1 Expires      SLA #2 Expires       Rep
Responds  │
│      │                   │                   │                     │
│
│      ▼                   ▼                   ▼                     ▼
│
│   T+0hrs              T+2hrs              T+3hrs               T+5hrs
│
│      │                   │                   │                     │
│
```

```
|       └────── 2hr SLA ──────┴────── 1hr SLA ──────┴────── 7 day SLA ─────┘
|

|             (Missed)            (Missed)              (Met @ 2hrs)
|

|
|

|  MQL_Calculated_TTC = 2hrs + 1hr + 2hrs = 5 hours
|

                                               └──────────────────────────────
                                               ───────────────────────────────
                                               ──────────────────────┘
```

SLA Cascade Structure

Order	SLA Window	Threshold	Triggered When
1	First	2 hours	MQL Opens (Last_MQL_Open_Date__c set)

2	Second	1 hour	First SLA missed
3	Third	7 days	Second SLA missed

How Time_To_Action_Calculated is Populated

- **If SLA is Met:** Actual elapsed time within that SLA window
- **If SLA is Missed:** Full SLA threshold value

This design ensures the sum always equals true elapsed time from MQL Open to response.

Technical Implementation

Components

Component	Type	Purpose
`TimetoActionTrackersHelper`	Apex Class	Calculates and updates MQL_Calculated_TTC__c

`TimetoActionTrackersTest`	Test Class	Unit tests for the helper
`LeanData__Time_To_Action_Tracker__c`	Managed Object	LeanData's tracker records
`Last_MQL_Open_Date__c`	Custom Field	Timestamp when MQL cycle began
`Last_MQL_Close_Date__c`	Custom Field	Timestamp when rep took action
`MQL_Calculated_TTC__c`	Custom Field	Calculated total response time

Trigger Flow

```
Time to Action Tracker Status Changes
```

```
          ▼
┌─────────────────────┐
│ Trigger Handler     │
└─────────────────────┘
          │
          ▼
┌─────────────────────┐
```

```
│ afterUpdateField  │

│ Updates()         │

└───────────────────┘

        │

        ▼

┌───────────────────┐

│ updateMQLCalcu-   │
```

```
│ latedTTC()        │

└───────────────────┘

        │

        ▼

┌───────────────────┐

│ Lead/Contact      │

│ MQL_Calculated_   │
```

```
| TTC__c Updated    |
└─────────────────────┘
```

Processing Logic

1. **Filter Criteria Check**
   - Status must be `'Met'` or `'Missed'`
   - Node name must be one of:
     - `'MQL Contacted Hold'`
     - `'Wait One Hour for MQL Follow Up'`
     - `'Continue MQL Hold'`
   - Status field must have changed (not just any field edit)
2. **MQL Window Calculation**
   - Uses `Last_MQL_Open_Date__c` and `Last_MQL_Close_Date__c` from parent record
   - Fallback logic if dates are null:
     - Open Date fallback: `Close Date - 7 days` or `Today - 7 days`
     - Close Date fallback: `Open Date + 7 days` or `Today + 7 days`
3. **Tracker Aggregation**
   - Queries ALL trackers for the Lead/Contact
   - Filters to only those with `CreatedDate` within the MQL window
   - Sums `LeanData__Time_To_Action_Calculated__c`
4. **Field Update**
   - Writes sum to `MQL_Calculated_TTC__c` on parent Lead/Contact

LeanData Configuration

Required Node Names

The Apex class looks for specific node names in LeanData FlowBuilder. Ensure your MQL routing flow uses these exact names:

```
public static final List<String> NODE_NAME_LIST = new List<String>{

    'Continue MQL Hold',

    'Wait One Hour for MQL Follow Up',

    'MQL Contacted Hold'
                                                };
```

⚠️ **Important:** If you rename nodes in LeanData, you must update this list in the Apex class or the trackers won't be processed.

Time to Action Node Configuration

Each Time to Action node in LeanData should be configured with:

- Appropriate SLA threshold (2hr, 1hr, 7 days)
- Action trigger tied to `Last_MQL_Close_Date__c` being populated
- Sequential flow so only one SLA is active at a time

Edge Cases & Considerations

Re-MQLs

If a Lead/Contact re-MQLs (new MQL cycle after previous one closed):

- New `Last_MQL_Open_Date__c` is set
- New trackers are created with new `CreatedDate`
- Window calculation isolates new cycle's trackers from historical ones
- `MQL_Calculated_TTC__c` reflects only the current/most recent cycle

Field Reference

Lead Fields

API Name	Type	Description
`Last_MQL_Open_Date__c`	DateTime	When the current MQL cycle began
`Last_MQL_Close_Date__c`	DateTime	When rep took action on MQL
`MQL_Calculated_TTC__c`	Number	Sum of response time in seconds

Contact Fields

API Name	Type	Description
`Last_MQL_Open_Date__c`	DateTime	When the current MQL cycle began
`Last_MQL_Close_Date__c`	DateTime	When rep took action on MQL
`MQL_Calculated_TTC__c`	Number	Sum of response time in seconds

Time to Action Tracker Fields (LeanData Managed)

API Name	Description
`LeanData__Lead__c`	Lookup to Lead
`LeanData__Contact__c`	Lookup to Contact
`LeanData__Node_Name__c`	Name of the FlowBuilder node

`LeanData__Status__c`	'Open', 'Met', or 'Missed'
`LeanData__Time_To_Action_Calculated__c`	Seconds elapsed or SLA threshold

Reporting

Useful Metrics

- **Average MQL Response Time:** `AVG(MQL_Calculated_TTC__c)` grouped by Owner, Team, or Region
- **SLA Compliance Rate:** Count of Trackers with Status = 'Met' vs 'Missed'
- **Response Time Distribution:** Bucket `MQL_Calculated_TTC__c` into ranges (0-2hr, 2-4hr, 4-8hr, 8hr+)

Sample SOQL

```
-- Average TTC by Lead Owner

SELECT
Owner
.
Name
```

```
,
AVG
(
MQL_Calculated_TTC__c
)
avgTTC
,
COUNT
(
Id
)
mqlCount
FROM
Lead
WHERE
Last_MQL_Close_Date__c
=
THIS_QUARTER
GROUP
BY
Owner
```

```
.

Name

ORDER
BY
AVG
(
MQL_Calculated_TTC__c
)
```

Troubleshooting

MQL_Calculated_TTC__c Not Updating

1. Verify tracker status changed to 'Met' or 'Missed'
2. Confirm node name matches one in `NODE_NAME_LIST`
3. Check that `Last_MQL_Open_Date__c` is populated on the Lead/Contact
4. Review Apex debug logs for errors in `TimetoActionTrackersHelper`

Incorrect Values

1. Check if trackers from previous MQL cycles are being included (window calculation issue)
2. Verify `LeanData__Time_To_Action_Calculated__c` values on individual trackers
3. Confirm no duplicate trackers exist for the same SLA window